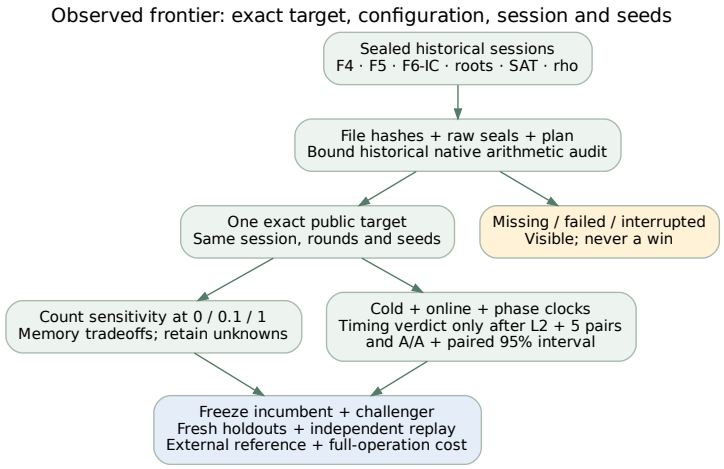
Backend and algorithm frontier

Initial integration and validation — 17 October 2026

Delivered. A standalone native Rust report imports 17 historical sessions into 626 exact-target backend entries, retaining 170 planned cells without measurements. F4 matrix, inherited F4, F5 matrix, F6-IC, custom root finding, SAT and reference methods share one view. No solver was run or retimed.

The incumbent to improve. The IC recipe frontier identifies nondominated IC configurations on the same exact target, session, rounds and seeds at all three sensitivity weights (0, 0.1 and 1 GAE per native operation). The cross-method frontier and memory tradeoffs remain separate. Raw solver counters retain their units; SAT conflict and learned-clause observations keep their original JSON source pointer. A stage gain is not a full-operation gain.



Coverage is evidence, not a study name. Fields of size 2^3, 2^4, 2^5, 2^{51} and 2^{83} have no imported measurements. Degree 53 does have observations, with their audit limits visible. The study named `ecbench_m83_rho_20261005` supplies calibration at degrees 53, 59 and 61; it does not establish degree-83 coverage. Interrupted degree-31 plans stay incomplete. The exact E_0 degree-83 scaling gate remains open.

Timing and validation. No timing winner is admitted by this panel's complete L2, five-pair, same-target A/A and paired-interval gates. The table shows descriptive cold solve, online, process and phase clocks. Native tests, lint, read-only freshness checks and desktop/mobile browser checks accompany the implementation. Historical arithmetic audits must bind the exact source files; this report performs no fresh arithmetic replay.

Existing validation gates. The root library on base `c6d64436` fails compilation (25 ecbench build errors; 26 library-test compile errors), in unchanged integration files. The existing site suite also has progress-JSON/panel-index failures and a shallow-input read limitation. These are recorded in the Markdown report and validation logs, not waived. The new report builds independently. No global-fastest, external-tool, asymptotic or ledger-promotion claim is made.

Reproduction and sources. See `docs/ecbench/BACKEND_FRONTIER.md`, `docs/backend-leaderboard/manifest.json` and `tools/backend-leaderboard/`. The JSON records source and producer-input hashes. Diagram source: `flow.dot`. Full commands, requirement status, limits and validation receipts: `REPORT.md` and `validation/`.